

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

GreenLedger: Development of a Role-Based Secure Records and Payment Management System for Memorial Park Services

In partial fulfillment for the requirements in Capstone 1 and Research Subject

ARDIENTE, JENELYN B. BSIT 3-4

BORAQUIT, APRIL JOHN E. BSIT 3-4

CABAÑAS, FRANCIS C. BSIT 3-4

CASIÑO, NOMER JOHN A. BSIT 3-4

CATALOGO, FRANCIS LEE R. BSIT 3-4

PARREÑO, MEL JOHN M. BSIT 3-4

Cyber Security

Second Semester

2026-2027

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In modern organizations, information serves as one of the most valuable resources in supporting daily operations, decision-making, and service delivery. Institutions that manage large amounts of client information and financial transactions require reliable systems to ensure that records are properly organized, secured, and easily accessible. Efficient record management enables organizations to maintain accurate information, monitor transactions effectively, and improve operational performance. As businesses continue to adopt digital technologies, computerized information systems have become essential tools for improving efficiency, minimizing operational errors, and strengthening data security.

The continuous advancement of information technology has significantly transformed the way organizations manage records and financial processes. Traditional methods such as paper-based documentation and spreadsheet monitoring are gradually being replaced by digital systems capable of storing, processing, and retrieving information more efficiently. Computerized management systems provide faster access to records, improve transaction monitoring, reduce data redundancy, and enhance the reliability of organizational information. In addition, centralized systems allow organizations to improve productivity while maintaining better control over confidential and sensitive records.

In the memorial park service industry, organizations manage various forms of important information, including client records, burial lot ownership details, contracts, payment transactions, and service-related documents. These records are essential in maintaining accurate operational procedures and ensuring financial accountability within the organization. Since memorial park services commonly involve long-term client transactions and installment-based payment arrangements, maintaining organized and secure records is necessary to prevent disputes, payment inconsistencies, and data loss.

Despite the advantages offered by digital technologies, some memorial park service organizations still rely on manual or semi-digital methods such as paper-based records, logbooks, and spreadsheet applications for managing operational and financial data. While these methods may initially appear manageable, they often become inefficient as the volume of records and transactions increases. Manual processes require significant time and effort for encoding, updating, searching, and verifying information. Retrieving client records and payment histories may become difficult and time-consuming, particularly when records are incomplete, unorganized, or inconsistently maintained.

Manual record management also increases the possibility of human errors, including incorrect data entry, duplication of records, incomplete transaction monitoring, and miscalculation of

balances. Physical records are vulnerable to deterioration, accidental damage, and loss due to improper handling and environmental factors. Similarly, spreadsheet-based systems may lack sufficient validation and monitoring mechanisms, making them prone to accidental modification or deletion of important data. These issues may negatively affect organizational efficiency, reliability, and the overall quality of service provided to clients.

Another major concern in organizational record management is data security. Sensitive client and financial information must be protected against unauthorized access, modification, and misuse. In systems without proper access control mechanisms, employees may gain unrestricted access to confidential records beyond their assigned responsibilities. The absence of monitoring and auditing features also makes it difficult to identify which users accessed or modified specific information within the system. These security weaknesses may compromise data integrity, accountability, and confidentiality within the organization.

Cybersecurity has become an important aspect of modern information systems, particularly for organizations that handle personal and financial information. Secure systems implement protective mechanisms such as authentication, role-based access control, activity logging, input validation, password hashing, and backup systems to minimize security risks and help improve data protection. Authentication mechanisms verify user identities before granting access to the system, while role-based access control limits system functions according to the responsibilities of authorized users. Activity logs allow administrators to monitor user actions and maintain accountability within the organization. In addition, backup systems help organizations recover important records in cases of hardware failure, accidental deletion, or cybersecurity-related incidents.

One widely used approach in securing information systems is Role-Based Access Control (RBAC). RBAC restricts system access by assigning permissions according to the roles and responsibilities of users within an organization. Instead of allowing all users to access the same information and functions, the system only permits authorized operations based on assigned user roles. For example, accounting personnel may access payment-related records, while administrative users manage user accounts and system configurations. This approach strengthens system security, minimizes unauthorized access, and improves the management of sensitive organizational information.

To address these operational and security challenges, organizations require an integrated digital system capable of managing records, monitoring financial transactions, and enforcing controlled access to sensitive information. A secure records and payment management system can improve organizational efficiency by automating record storage, simplifying transaction monitoring, reducing human errors, and strengthening information security. Through centralized data management, organizations can retrieve records more efficiently while maintaining accurate and reliable documentation of transactions and client information.

Therefore, this study proposes the development of **GreenLedger: Development of a Role-Based Secure Records and Payment Management System for Memorial Park Services**. The proposed system aims to provide a secure, organized, and efficient platform for

managing client records, memorial lot information, and payment transactions. The system will implement role-based access control, authentication mechanisms, activity logging, and backup features to ensure controlled access, accountability, and data protection among authorized users. Furthermore, the proposed system seeks to improve operational efficiency, strengthen information security, and support reliable record management practices within memorial park service organizations.

1.2 Statement of the Problem

Organizations that rely on manual or semi-digital systems often experience difficulties in maintaining accurate, organized, and secure records. In memorial park service organizations, proper management of client information, burial lot records, and payment transactions is essential to ensure operational efficiency and financial accountability. However, based on the conducted interview and observation at Greengarden Inc., several challenges were identified in the organization's existing records management practices.

The organization currently depends on manual documentation and spreadsheet-based monitoring for handling client records and financial transactions. As the volume of records and transactions increases, the process becomes more difficult to manage efficiently. This situation may result in delays in updating records, inconsistencies in stored information, difficulties in retrieving data, and increased risks of human error. Furthermore, the absence of sufficient security and monitoring mechanisms may expose sensitive client and financial information to unauthorized access, modification, and possible data loss.

This study seeks to address the following problems:

1. The current record management process is manual or spreadsheet-based, making data organization, retrieval, and updating time-consuming and prone to human errors.
2. Payment transactions are not monitored efficiently, which may lead to delayed updates, inaccurate balances, and financial inconsistencies.
3. Sensitive client and financial records are not fully protected due to limited access control and insufficient security mechanisms.
4. The existing process lacks activity monitoring and audit logging, making it difficult to trace user actions and record modifications.
5. Backup and recovery mechanisms are insufficient, increasing the risk of data loss during hardware failure, accidental deletion, or cybersecurity incidents.
6. The organization lacks a centralized and integrated system capable of securely managing records, payment transactions, and user access within a single platform.

1.3 Objectives of the Study

1.3.1 General Objectives

The general objective of this study is to design and develop GreenLedger: A Role-Based Secure Records and Payment Management System for Greengarden Inc. that will improve record organization, transaction monitoring, controlled access, and data security through the implementation of a secure web-based management system.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze the existing manual and semi-digital record management practices of Greengarden Inc. to identify operational and security-related limitations.
2. Design and develop a secure web-based system for managing client records, burial lot information, and payment transactions.
3. Implement Role-Based Access Control (RBAC) to restrict system functions and data accessibility according to user roles and responsibilities.
4. Develop a login and authentication mechanism that ensures only authorized users can access the system.
5. Implement payment tracking features capable of recording and monitoring financial transactions accurately and efficiently.
6. Integrate activity logs to monitor user actions, improve accountability, and support auditing processes within the organization.
7. Implement cybersecurity features such as password hashing, input validation, office-restricted access, and backup systems to enhance data protection and system security.
8. Evaluate the system based on functionality, usability, security, efficiency, and reliability.

1.4 Scope and Delimitation

1.4.1 Scope of the Study

This study focuses on the design, development, and evaluation of GreenLedger: A Role-Based Secure Records and Payment Management System for Memorial Park Services for Greengarden Inc. The proposed system is a web-based application intended to improve the organization's current process of managing client records, memorial lot information, payment transactions, and administrative monitoring. The study aims to provide a more organized, secure, and efficient method of handling operational and financial records compared to the organization's existing manual and spreadsheet-based processes.

The proposed system is designed to operate within the office environment of Greengarden Inc. and will be accessible only to authorized personnel through a secure login system. The system supports multiple user roles, including administrators, accounting staff, and general staff, with each role having specific permissions and access limitations through the implementation of Role-Based Access Control (RBAC). This mechanism ensures that users can only access system functions and information relevant to their assigned responsibilities.

The system includes a user authentication module that verifies user identities before access is granted to the platform. Authentication features include username and password verification, secure login sessions, and password protection mechanisms provided through the Django framework. The system also incorporates password hashing to protect user credentials from unauthorized disclosure within the database.

GreenLedger includes modules for client information management and memorial lot record management. Authorized personnel may add, update, search, and retrieve client records and memorial lot information using the system interface. The system is intended to reduce difficulties commonly associated with manual documentation, such as misplaced records, duplicated information, delayed retrieval of documents, and inconsistencies in record updating.

The study also covers the implementation of a payment tracking and monitoring module. This module allows authorized accounting personnel to record client payments, monitor payment balances, track payment schedules, and update transaction records in real time. The system is designed to improve financial monitoring accuracy and reduce human errors associated with manual computation and spreadsheet-based tracking.

To strengthen accountability and monitoring, the proposed system includes activity logging and audit trail functionality. User actions performed within the system, such as login attempts, record modifications, payment updates, and administrative changes, are automatically recorded in activity logs. These logs help administrators monitor system usage, review user activities, and identify unauthorized or suspicious actions when necessary.

The study further includes the implementation of security and data protection mechanisms appropriate to the nature of the system. These mechanisms include Role-Based Access Control

(RBAC), password hashing, authentication processes, input validation, and office-restricted access. Input validation and sanitization are implemented to reduce the risk of invalid or malicious data entries that may affect the integrity of the system database.

In terms of data protection and recovery, the study incorporates backup mechanisms intended to reduce the risk of data loss caused by accidental deletion, hardware malfunction, or system-related incidents. The system includes local backup generation of records and scheduled cloud-based backup storage for additional recovery support. Cloud backup copies are automatically managed and periodically removed after a specified retention period to optimize storage usage and maintain system performance efficiency.

The study also includes the deployment and evaluation of the system prototype within the operational environment of Greengarden Inc. The developed system will be implemented for a trial period to allow users to evaluate its functionality, usability, efficiency, and security. Feedback gathered from administrators and staff during the implementation period will be used as part of the system evaluation and improvement process before the final system assessment and defense.

Furthermore, the study focuses on the application of cybersecurity principles in records and payment management systems. As a cybersecurity-related capstone project, the researchers aim to demonstrate the practical implementation of secure authentication, access control, monitoring, and data protection mechanisms within a real-world organizational setting.

1.4.2 Delimitation of the Study

This study is limited only to the development and implementation of a prototype web-based records and payment management system for the internal operations of Greengarden Inc. The proposed system is intended solely for organizational use and does not support public access, client self-service portals, or external online transactions.

The study does not include integration with external payment gateways, online banking systems, digital wallet platforms, or automated online payment processing services. Payment transactions covered in the system are limited only to internal payment recording and monitoring performed by authorized personnel within the organization.

The proposed system is designed for a single organizational environment only and does not support multi-branch deployment, distributed database systems, or enterprise-scale infrastructure. The system will operate primarily within the office network environment of Greengarden Inc. and is not intended for nationwide or large-scale commercial implementation.

The study is limited to a web-based platform and does not include mobile application development for Android or iOS devices. Access to the system will primarily be performed through desktop or laptop computers connected to the organization's designated network environment.

Although the study incorporates several cybersecurity mechanisms such as authentication, password hashing, Role-Based Access Control (RBAC), activity logging, and input validation, the system does not implement highly advanced enterprise-level cybersecurity technologies. Features such as biometric authentication, artificial intelligence-based threat detection, intrusion prevention systems, blockchain-based security, penetration testing automation, and large-scale cloud-native security infrastructures are beyond the scope of the study.

The cloud backup functionality included in the study is limited to scheduled backup storage intended for recovery support and data redundancy. Due to storage limitations and performance considerations, cloud backup copies are managed with limited retention periods and are not intended to function as permanent large-scale archival storage systems.

The evaluation of the proposed system is limited to functionality, usability, efficiency, reliability, and security within the context of the organization's operational requirements. The study does not include long-term enterprise performance testing, large-scale concurrent user stress testing, or commercial-grade cybersecurity certification assessments.

Lastly, the system is developed primarily for academic and research purposes as part of the researchers' capstone project requirement in Cyber Security. While the system may assist the organization in improving its operations, the study does not guarantee complete elimination of operational risks, cybersecurity threats, or human errors within the organization.

1.5 Significance of the Study

The proposed study is significant because it aims to improve the efficiency, accuracy, and security of record and payment management processes within Greengarden Inc. The development of GreenLedger may provide several benefits to the following groups:

- **Administrators** –The system will assist administrators in managing organizational records, monitoring user activities, and enforcing controlled access to sensitive information. Activity logs and role-based permissions will improve accountability and supervision within the organization.
- **Accounting and Financial Staff** – The system will provide a more organized and efficient method of recording, updating, and monitoring payment transactions. This may reduce financial errors, improve reporting accuracy, and simplify transaction tracking.
- **General Staff and End Users** – Authorized staff members will benefit from faster record retrieval, improved organization of client information, and simplified workflow processes through a centralized digital platform.
- **Clients** – The proposed system may help ensure that client records and payment information are properly managed, secured, and accurately maintained, improving reliability and service quality.⁴⁷
- **Greengarden Inc.** – The organization may benefit from improved operational efficiency, reduced manual workload, enhanced record security, and more reliable management of client and financial information.
- **Cybersecurity Students and Researchers** – The study may serve as a practical reference for implementing cybersecurity concepts such as Role-Based Access Control (RBAC), authentication, activity logging, password hashing, and secure system design within a real-world management system.
- **Future Researchers and Developers** – future researchers and system developers may use this study as a reference in developing similar secure information systems involving records management, payment tracking, and access control mechanisms. By enabling effective supervision, monitoring, and auditing of records and financial transactions.

1.6 Definition of Terms

- **Activity Logs** – Records generated by the system that track user actions, login activities, and modifications performed within the application for monitoring and auditing purposes.
- **Authentication** – The process of verifying the identity of a user before granting access to the system.
- **Backup System** – A security mechanism that creates copies of important records and system data to prevent permanent loss during system failures or cybersecurity incidents.
- **Burial Lot Management** – The process of organizing and maintaining information related to memorial lot ownership, status, and allocation.
- **Client Records** – Personal and transaction-related information of clients stored and managed within the system.
- **Cybersecurity** – The practice of protecting systems, networks, and digital information from unauthorized access, attacks, and data breaches.
- **GreenLedger** – The proposed role-based secure records and payment management system developed for Greengarden Inc.
- **Input Validation** – A security process used to verify and sanitize user inputs to prevent invalid or malicious data from entering the system.
- **Memorial Park Services** – Services related to memorial lot management, interment processes, and client transaction handling provided by Greengarden Inc.
- **Office-Restricted Access** – A security feature that limits system accessibility to authorized users within the organization's office network or designated environment.
- **Password Hashing** – A security technique that converts passwords into encrypted representations before storing them in the database to protect user credentials.
- **Payment Tracking** – The process of recording, updating, and monitoring client payment transactions within the system.

- **Role-Based Access Control (RBAC)** – A security mechanism that restricts system access and permissions according to the assigned roles and responsibilities of users.
- **Secure System** – A system designed with protective mechanisms to ensure confidentiality, integrity, and controlled access to sensitive information.
- **Web-Based System** – An application that operates through a web browser and allows users to access system functions through a network connection.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies relevant to the development of GreenLedger: A Role-Based Secure Records and Payment Management System for Memorial Park Services. The reviewed literature includes concepts, technologies, methodologies, and cybersecurity practices associated with records management systems, payment monitoring systems, authentication mechanisms, role-based access control, audit logging, database management systems, and web-based organizational systems. These concepts provide the theoretical and technical foundation necessary for the development of the proposed system.

The chapter also presents foreign and local studies related to secure records management, electronic payment monitoring, organizational information systems, and cybersecurity implementation. The reviewed studies provide insights into current technologies, security mechanisms, and system development practices used in organizational environments. Additionally, these studies help identify research gaps and existing challenges related to records management and information security.

The purpose of this review is to establish the relevance of the proposed study by examining existing concepts, systems, and research related to secure records and payment management. Through this chapter, the researchers aim to support the development of GreenLedger by identifying applicable technologies, security practices, and organizational solutions relevant to memorial park service operations.

2.1 Related Literature

2.1.1 Information Systems

Information systems are essential components of modern organizations because they support operational processes, communication, data management, and decision-making activities. According to Kenneth Laudon and Jane Laudon (2022), an information system is composed of interconnected components that collect, process, store, and distribute information to support coordination, management, and organizational control. Information systems help organizations improve efficiency by

automating processes and organizing large amounts of data into manageable digital formats.

Organizations use information systems to improve operational performance and reduce inefficiencies associated with manual processes. Traditional methods of storing records through paper documents and spreadsheets often lead to duplication, delays in retrieval, and inconsistent monitoring of organizational data. Through computerized information systems, organizations can centralize records, simplify transactions, and improve communication among departments and personnel.

Information systems also contribute to organizational accountability by maintaining accurate and updated records of transactions and activities. Automated systems reduce repetitive tasks and minimize human errors commonly associated with manual encoding and monitoring processes. In addition, centralized information systems improve accessibility because authorized users can retrieve records more efficiently within a controlled environment.

In organizations handling sensitive records and financial information, information systems also support data security and confidentiality. Modern systems implement security mechanisms such as authentication, access control, encryption, and audit logging to protect information from unauthorized access and modification. These security measures help organizations maintain data integrity and operational reliability.

The proposed GreenLedger system is related to this literature because it functions as an organizational information system intended to manage client records, memorial lot information, and payment transactions securely and efficiently. The system aims to improve operational processes by replacing manual and spreadsheet-based procedures with a centralized digital platform.

Reference

Laudon, K. C., & Laudon, J. P. (2022). *Management Information Systems: Managing the Digital Firm* (17th ed.). Pearson Education.

2.1.2 Web-Based Systems

Web-based systems are information systems that operate through web technologies and network-connected environments. According to Roger Pressman and Maxim (2019), web-based applications allow organizations to deliver services, process

transactions, and manage organizational information through centralized platforms accessible using web browsers. These systems are widely used because they simplify accessibility, maintenance, and management of organizational data.

Unlike traditional standalone applications, web-based systems store data and process requests through centralized servers and databases. Authorized users can access system functions using devices connected within the organizational network. This setup improves collaboration and allows multiple users to interact with the same database environment simultaneously.

Web-based systems provide several advantages to organizations, including centralized record storage, faster retrieval of information, improved transaction processing, and simplified system maintenance. Because data is stored centrally, organizations can monitor updates and activities more efficiently. In addition, system updates and modifications can be implemented directly on the server without requiring manual installation on multiple devices.

Security is also an important consideration in web-based systems. Since organizational information is accessed digitally, web systems implement authentication mechanisms, role-based permissions, session management, and input validation to protect data against unauthorized access and malicious activities. These security measures are essential for organizations handling confidential client and financial information.

The proposed GreenLedger system is related to this literature because it is designed as a web-based records and payment management system intended for use within the office environment of Greengarden Inc. Through centralized web technologies, the system aims to improve accessibility, transaction monitoring, and secure management of organizational records.

Reference

Pressman, R. S., & Maxim, B. R. (2019). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.

2.1.3 Records Management Systems

Records management systems are designed to organize, store, retrieve, and maintain organizational records efficiently and securely. According to the International Organization for Standardization through ISO 15489, records management involves the

systematic control of the creation, storage, use, maintenance, and disposal of records to support organizational operations and accountability. Effective records management ensures that important documents and transaction records remain accurate, accessible, and protected throughout their lifecycle.

Organizations handling large amounts of operational and financial information require structured records management systems to maintain efficiency and reliability. Traditional paper-based systems often create difficulties in organizing files, locating records, and monitoring transactions. As the volume of records increases, manual methods become more time-consuming and vulnerable to human errors, duplication, and document loss.

Digital records management systems improve organizational efficiency by automating record storage and retrieval processes. Through computerized systems, organizations can categorize records, search information quickly, and update transactions more accurately. Electronic systems also reduce physical storage requirements and improve long-term preservation of organizational data.

Another important advantage of records management systems is data security. Modern systems implement access control mechanisms, authentication procedures, and activity monitoring to ensure that only authorized users can access confidential records. Backup and recovery mechanisms also help organizations restore important data during hardware failures or unexpected incidents.

The proposed GreenLedger system is related to this literature because it focuses on managing client records, burial lot information, and payment transactions within a centralized records management environment. The system aims to improve record organization, reduce manual errors, and strengthen the security of organizational information.

Reference

International Organization for Standardization. (2016). ISO 15489-1: Information and documentation — Records management. ISO.

2.1.4 Payment Management Systems

Payment management systems are computerized systems used to monitor, process, record, and organize financial transactions within organizations. These systems help businesses maintain accurate financial documentation, monitor balances, and improve

transaction efficiency. According to Oracle Corporation (2022), digital payment and financial management systems improve transaction accuracy by automating payment recording and reducing manual processing errors.

Organizations that manage installment-based transactions require effective payment monitoring systems to maintain accurate financial records. Manual monitoring methods often lead to delayed updates, incorrect balances, incomplete payment histories, and inconsistencies in financial reporting. Spreadsheet-based monitoring systems may also become difficult to maintain as transaction volumes increase over time.

Computerized payment management systems allow organizations to record payments in real time and automatically compute balances, dues, and transaction histories. These systems improve organizational accountability by maintaining accurate records of payment activities and reducing the possibility of human miscalculations. In addition, payment systems simplify financial monitoring because authorized personnel can retrieve transaction histories quickly and efficiently.

Security is also essential in payment management systems because financial information is considered sensitive organizational data. Modern systems implement authentication, role-based permissions, activity logging, and backup systems to protect transaction records from unauthorized access and accidental modification. These mechanisms help ensure financial integrity and accountability within organizations.

The proposed GreenLedger system is related to this literature because it includes a payment tracking and monitoring module intended to manage installment payments, balances, and transaction histories securely and accurately for Greengarden Inc.

Reference

Oracle Corporation. (2022). Financial Management and Payment Systems Overview. Oracle Documentation.

2.1.5 Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a security mechanism that restricts system access according to the roles and responsibilities assigned to users within an organization. According to Ravi Sandhu, Coyne, Feinstein, and Youman (1996), RBAC simplifies access management by assigning permissions to organizational roles instead

of assigning permissions individually to every user. This approach improves security, accountability, and administrative efficiency in multi-user systems.

In traditional systems without proper access restrictions, users may gain unnecessary access to confidential organizational information. Excessive access privileges increase the risk of unauthorized viewing, modification, or misuse of sensitive records. RBAC addresses this issue by ensuring that users can only access system functions relevant to their assigned responsibilities.

RBAC is widely used in organizational information systems because it supports controlled access management and strengthens data security. For example, administrators may manage user accounts and system settings, while accounting personnel are limited to payment monitoring and financial records. General staff members may only access operational records relevant to their assigned tasks. Through controlled permissions, organizations reduce security risks and improve operational accountability.

Another advantage of RBAC is simplified system administration. Since permissions are assigned according to roles, administrators can manage access privileges more efficiently without configuring permissions individually for every user. This approach becomes especially useful in organizations with multiple departments and varying access requirements.

The proposed GreenLedger system implements RBAC to ensure that only authorized personnel can access sensitive records and system functions. The system assigns permissions based on user roles such as administrator, accounting staff, and general staff to improve information security and operational control.

Reference

Sandhu, R., Coyne, E., Feinstein, H., & Youman, C. (1996). Role-Based Access Control Models. *IEEE Computer*, 29(2), 38–47.

2.1.6 Authentication Systems

Authentication systems are security mechanisms used to verify the identity of users before granting access to information systems. Authentication ensures that only authorized individuals can access protected resources, records, and organizational

functions. According to National Institute of Standards and Technology (2020), authentication is one of the fundamental components of information security because it protects systems against unauthorized access and identity misuse.

Most authentication systems require users to provide credentials such as usernames and passwords before entering the system. Advanced authentication methods may also include multi-factor authentication, biometric verification, and token-based security. However, username and password authentication remains one of the most commonly used methods in organizational systems because of its simplicity and practicality.

Weak authentication systems expose organizations to risks such as unauthorized access, identity theft, and data breaches. If user accounts are not properly protected, unauthorized individuals may gain access to confidential information and modify important records. Therefore, organizations implement authentication mechanisms together with password protection, session management, and access control policies to strengthen system security.

Authentication systems also improve accountability because user activities can be associated with verified identities. Once users log into the system, their actions can be recorded through activity logs and audit trails. This helps organizations monitor system usage and investigate unauthorized activities when necessary.

The proposed GreenLedger system uses authentication mechanisms to verify users before granting access to organizational records and payment transactions. Through secure login procedures and controlled access management, the system aims to protect sensitive information from unauthorized access.

Reference

National Institute of Standards and Technology. (2020). Digital Identity Guidelines. NIST Special Publication 800-63.

2.1.6 Password Hashing

Password hashing is a security technique used to protect user passwords by converting them into encrypted representations before storing them in a database. Instead of saving actual passwords in plain text, systems store hashed versions generated through mathematical algorithms. According to OWASP Foundation (2023), password hashing is

an essential security practice that helps protect user credentials from unauthorized access and database breaches.

When users create passwords, the system processes the input through a hashing algorithm that generates a unique fixed-length value called a hash. During login, the entered password is hashed again and compared to the stored hash value. Since hashing is a one-way process, the original password cannot easily be retrieved from the stored hash. This mechanism improves confidentiality and reduces the risk of credential exposure.

Modern systems use secure hashing algorithms such as bcrypt, Argon2, and PBKDF2 because they are designed to resist password cracking techniques. Weak or outdated hashing methods may expose systems to brute-force attacks and credential theft. Therefore, organizations handling sensitive information implement stronger password protection mechanisms to improve account security.

Password hashing also contributes to organizational cybersecurity by minimizing the impact of database compromise incidents. Even if attackers gain access to the database, properly hashed passwords remain difficult to decode. Combined with authentication systems and access control mechanisms, password hashing strengthens overall system security.

The proposed GreenLedger system incorporates password hashing to protect user credentials and improve the security of organizational accounts. Through secure password storage mechanisms, the system aims to reduce risks associated with unauthorized account access and credential exposure.

Reference

OWASP Foundation. (2023). Password Storage Cheat Sheet. OWASP.

2.1.7 Input Validation

Input validation is a security process used to verify and filter user-provided data before it is processed by an information system. According to the OWASP Foundation (2023), input validation helps protect systems against malicious data, invalid inputs, and common web application attacks such as SQL injection and cross-site scripting.

Organizations use input validation to ensure that information entered into the system follows required formats and rules. For example, numeric fields should only accept numbers, while email fields must follow valid email structures. By validating user input, systems minimize errors, maintain data consistency, and prevent unauthorized manipulation of information.

Without proper input validation, systems become vulnerable to malicious attacks and corrupted data entries. Attackers may exploit weak validation mechanisms to inject harmful commands into databases or manipulate system behavior. These vulnerabilities may compromise organizational records, disrupt operations, and expose sensitive information.

Input validation also improves data quality and reliability within organizations. By ensuring that users provide correct and properly formatted information, organizations maintain cleaner databases and reduce inconsistencies in stored records. This contributes to better transaction monitoring and more accurate reporting processes.

The proposed GreenLedger system implements input validation to ensure that records, payment details, and user information entered into the system follow proper formats and security standards. This mechanism helps improve data integrity and strengthen system security.

Reference

OWASP Foundation. (2023). Input Validation Cheat Sheet. OWASP.

2.1.8 Audit Logs

Audit logs are records of system activities used to monitor user actions, system events, and transaction histories within an information system. According to National Institute of Standards and Technology (2018), audit logging supports accountability, monitoring, and incident investigation by maintaining records of activities performed within organizational systems.

Organizations implement audit logs to track actions such as logins, record modifications, payment updates, and administrative activities. These logs help administrators identify who accessed the system, what actions were performed, and when the activities occurred. Audit trails improve transparency and accountability because user activities can be reviewed and monitored systematically.

Audit logs also contribute to cybersecurity and organizational security monitoring. Through activity records, administrators can detect suspicious behaviors, unauthorized access attempts, and unusual system activities. In the event of security incidents, audit logs provide evidence useful for investigation and recovery processes.

Without proper activity logging, organizations may experience difficulties in identifying the source of errors, unauthorized changes, or security breaches. The absence of monitoring mechanisms reduces accountability and limits the organization's ability to investigate incidents effectively.

The proposed GreenLedger system includes audit logging features to monitor user activities and maintain accountability within the organization. Through activity logs, administrators can track system usage, review transaction histories, and monitor modifications made by authorized users.

Reference

National Institute of Standards and Technology. (2018). Guide to Computer Security Log Management. NIST Special Publication 800-92.

2.1.8 Backup and Recovery Systems

Backup and recovery systems are mechanisms used to create copies of organizational data and restore information in case of data loss, hardware failure, accidental deletion, or cybersecurity incidents. According to National Institute of Standards and Technology (2021), backup systems are essential components of information security and organizational continuity because they help maintain data availability and operational recovery during unexpected events.

Organizations handling sensitive records and financial transactions require reliable backup mechanisms to protect important organizational information. Data loss incidents caused by hardware malfunction, human error, malware attacks, or natural disasters may significantly disrupt organizational operations if no backup systems are available.

Backup systems create secondary copies of records stored separately from the primary database. These backups may be stored locally, externally, or through cloud-based services depending on organizational requirements. Recovery mechanisms allow organizations to restore lost or damaged information using backup copies.

Regular backup schedules improve organizational reliability because updated copies of records remain available when recovery becomes necessary. However, organizations must also manage storage limitations and system performance considerations when implementing backup systems. Some organizations use scheduled deletion or backup rotation mechanisms to manage storage efficiently while maintaining data protection.

The proposed GreenLedger system incorporates backup and recovery mechanisms to protect client records and payment transactions from data loss. The system includes local backup files and scheduled cloud backup processes intended to improve organizational data security and recovery capability.

Reference

National Institute of Standards and Technology. (2021). *Contingency Planning Guide for Federal Information Systems*. NIST Special Publication 800-34.

2.1.9 Agile Methodology

Agile methodology is a software development approach focused on iterative development, collaboration, adaptability, and continuous improvement throughout the system development process. According to Kent Beck et al. (2001), Agile development emphasizes flexibility and incremental progress through continuous feedback and collaboration among developers and stakeholders.

Traditional software development methods often follow rigid sequential processes that may become difficult to adjust when requirements change during development. Agile methodology addresses this limitation by dividing development into smaller iterative phases called sprints or cycles. Each cycle focuses on developing, testing, and improving specific system features.

Agile development improves communication between developers and users because feedback is continuously gathered throughout the project. This allows developers to

identify issues early and implement necessary improvements during development rather than waiting until the final stages of the project.

Another advantage of Agile methodology is improved adaptability. Organizations and developers can modify system requirements and priorities based on operational needs and evaluation results. This approach helps ensure that the developed system remains aligned with user requirements and organizational objectives.

The proposed GreenLedger system uses Agile methodology during system development to support continuous improvement, iterative testing, and gradual enhancement of system features. Through Agile practices, the researchers aim to improve usability, functionality, and security throughout the development process.

Reference

Beck, K., Beedle, M., Van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., et al. (2001). *Manifesto for Agile Software Development*. Agile Alliance.

2.1.10 Cybersecurity in Organizations

Cybersecurity in organizations involves the protection of information systems, networks, databases, and digital resources against unauthorized access, cyberattacks, data breaches, and operational disruptions. According to the Cybersecurity and Infrastructure Security Agency (2023), cybersecurity practices are essential for maintaining the confidentiality, integrity, and availability of organizational information.

As organizations increasingly rely on digital systems, cybersecurity has become an important aspect of operational management. Organizations handling sensitive records and financial information face risks such as malware attacks, phishing, unauthorized access, data theft, and system vulnerabilities. Without proper security mechanisms, organizational operations and confidential information may become compromised.

Modern cybersecurity practices include authentication systems, access control, password protection, activity monitoring, encryption, backup systems, and security awareness procedures. These mechanisms help organizations protect digital assets and reduce vulnerabilities associated with computerized operations. Security policies and monitoring systems also improve organizational accountability and incident response capability.

Cybersecurity also supports data integrity by preventing unauthorized modification of records and transactions. In systems handling financial data, maintaining accurate and secure transaction histories is essential for operational reliability and accountability. Organizations implement layered security measures to minimize risks and improve protection against evolving cyber threats.

The proposed GreenLedger system incorporates cybersecurity practices such as authentication, role-based access control, password hashing, input validation, audit logging, and backup mechanisms to improve organizational security and protect sensitive records and financial information.

Reference

Cybersecurity and Infrastructure Security Agency. (2023). *Cybersecurity Best Practices*. CISA.

2.2 Foreign Studies

2.2.1 RBAC Implementation Studies

Studies related to Role-Based Access Control (RBAC) emphasize the importance of restricting system access according to organizational roles and responsibilities. In modern information systems, RBAC is widely implemented to strengthen security, improve accountability, and minimize unauthorized access to sensitive organizational records.

A study conducted by Meriem Ouzzani, Hammoudeh, and Al-Fayoumi (2021) examined the implementation of Role-Based Access Control in web-based organizational systems. The researchers found that RBAC improves system security by limiting user access according to organizational responsibilities and reducing unauthorized access to confidential records. The study emphasized that RBAC simplifies permission management in multi-user environments and strengthens accountability within information systems.

Similarly, David Ferraiolo, Kuhn, and Chandramouli (2020) discussed how RBAC supports secure organizational systems by assigning permissions according to defined operational roles. The researchers concluded that RBAC improves security

administration and minimizes insider threats by preventing unnecessary access privileges and unauthorized system activities.

Both studies are relevant to the proposed GreenLedger system because the proposed study also implements role-based access mechanisms to control access to sensitive records, payment transactions, and administrative functions. Similar to the reviewed studies, the present system aims to improve organizational security and accountability by ensuring that only authorized users can access specific system functions and information.

References

- Ouzzani, M., Hammoudeh, M., & Al-Fayoumi, M. (2021). *Role-Based Access Control in Web Information Systems*. International Journal of Information Security.
- Ferraiolo, D., Kuhn, D., & Chandramouli, R. (2020). *Role-Based Access Control and Organizational Security*. NIST Publications.

2.2.2 Electronic Records Management Studies

Electronic records management studies emphasize the importance of digital systems in improving document organization, accessibility, and operational efficiency. Organizations transitioning from manual processes to digital platforms benefit from centralized storage, faster retrieval of records, and improved management of organizational information.

A study conducted by Julie McLeod (2021) explored the effectiveness of electronic records management systems in improving organizational efficiency and document retrieval processes. The study found that computerized record systems significantly reduce document loss, improve information accessibility, and simplify monitoring of organizational transactions. The researcher emphasized that digital systems improve control over organizational information and reduce inefficiencies associated with manual filing systems.

Similarly, Anne Gilliland (2022) examined digital records management practices in organizations transitioning from paper-based systems to electronic platforms. The study concluded that centralized digital records systems improve operational consistency, accessibility, and information security. The researcher also highlighted the importance of

implementing security controls and backup mechanisms within electronic records environments.

Both studies are related to the proposed GreenLedger system because the present study also focuses on improving records management through a centralized digital platform. Similar to the reviewed studies, the proposed system aims to improve accessibility, organization, and protection of organizational records and financial transactions.

References

McLeod, J. (2021). *Electronic Records Management and Organizational Efficiency*. Records Management Journal.

Gilliland, A. (2022). *Digital Records Management Practices in Modern Organizations*. Information and Management Review.

2.2.3 Payment Tracking System Studies

Payment tracking system studies focus on improving financial monitoring, transaction accuracy, and accountability through computerized systems. Digital payment systems help organizations reduce human errors, monitor balances efficiently, and improve financial reporting processes.

A study conducted by Ahmed Elragal (2021) examined the implementation of computerized payment monitoring systems in service-oriented organizations. The study found that digital payment systems improve transaction accuracy, reduce manual errors, and simplify financial monitoring processes. The researcher emphasized that automated systems provide real-time transaction updates and improve financial accountability within organizations.

The findings are related to the proposed GreenLedger system because the present study also focuses on implementing a secure payment tracking and transaction monitoring system. Similar to the reviewed study, the proposed system aims to improve financial monitoring, minimize transaction inconsistencies, and support reliable payment management processes.

Reference

Elragal, A. (2021). *Digital Payment Monitoring Systems in Service Organizations*. International Journal of Financial Technology.

2.2.4 Audit Logging Studies

Audit logging studies emphasize the importance of monitoring user activities and maintaining accountability within organizational systems. Audit logs help organizations track system activities, investigate incidents, and detect unauthorized access attempts.

A study conducted by Karen Kent and Souppaya (2021) examined the role of audit logging in organizational cybersecurity and system monitoring. The researchers explained that audit logs improve accountability by recording user activities, login attempts, transaction histories, and system modifications. The study emphasized that organizations implementing proper audit logging mechanisms are more capable of identifying security violations and maintaining operational transparency.

Similarly, a related study conducted by Sajjad Haider, Rahman, and Ali (2022) explored the effectiveness of activity logging systems in detecting unauthorized activities within organizational information systems. The researchers concluded that monitoring systems with detailed activity logs improve incident response capability and support faster investigation of security-related events. The study also highlighted that audit logs become more effective when integrated with authentication systems and access control mechanisms.

Both studies are relevant to the proposed GreenLedger system because the proposed study also incorporates audit logging and activity monitoring features to improve accountability and strengthen security management within the organization.

References

Kent, K., & Souppaya, M. (2021). *Guide to Computer Security Log Management*. NIST Special Publication 800-92.

Haider, S., Rahman, M., & Ali, K. (2022). *Activity Logging and Organizational Security Monitoring*. Journal of Information Security Research.

2.2.5 Web-Based Management System Studies

Web-based management system studies highlight the effectiveness of centralized online platforms in improving organizational operations, accessibility of information, and transaction management. Web systems allow authorized users to access organizational records efficiently while supporting collaboration and operational productivity.

A study conducted by James Hendler, Berners-Lee, and Miller (2021) examined the effectiveness of web-based management systems in improving organizational operations and accessibility of records. The researchers found that centralized web applications improve transaction processing, simplify information retrieval, and reduce operational inefficiencies associated with manual systems.

Similarly, Mohammed Alenezi (2023) evaluated web-based management systems used in service-oriented organizations. The study concluded that web-based platforms improve operational efficiency by automating transactions, organizing records centrally, and improving accessibility for authorized personnel. The researcher also emphasized the importance of integrating authentication and access control mechanisms in web systems handling sensitive information.

Both studies support the proposed GreenLedger system because the present study also develops a web-based management platform intended to improve organizational operations, records management, and transaction monitoring while maintaining secure access to sensitive information.

References

Hendler, J., Berners-Lee, T., & Miller, E. (2021). *Web-Based Information Systems and Organizational Efficiency*. International Journal of Web Engineering.

Alenezi, M. (2023). *Secure Web-Based Management Systems for Service Organizations*. International Journal of Computer Applications.

2.2.6 Secure Database System Studies

Secure database studies focus on protecting organizational information from unauthorized access, modification, and security threats. Databases storing sensitive records require security controls to maintain confidentiality, integrity, and availability of information.

A study conducted by Elisa Bertino (2020) discussed the importance of secure database systems in protecting organizational records and confidential information. The researcher explained that databases storing financial and personal information require access control, authentication, backup systems, and encryption support to minimize security risks.

Similarly, Rakesh Agrawal, Kiernan, and Xu (2021) examined secure database implementation strategies for organizational systems handling confidential records. The researchers concluded that proper database management and security policies improve data reliability and reduce risks associated with unauthorized access and accidental data modification.

Both studies are relevant to the proposed GreenLedger system because the proposed study also relies on a centralized database system for storing client records, payment transactions, and activity logs securely. Similar to the reviewed studies, the proposed system prioritizes secure data management and controlled access to sensitive information.

References

Bertino, E. (2020). *Database Security and Organizational Information Protection*. Information Systems Security Journal.

Agrawal, R., Kiernan, J., & Xu, Y. (2021). *Secure Database Systems in Organizational Environments*. Journal of Data Security.

2.2.7 Cybersecurity Implementation Studies

Cybersecurity implementation studies emphasize the importance of integrating security mechanisms within organizational systems to protect digital records, transactions, and confidential information against cyber threats and unauthorized access.

A study conducted by Bruce Schneier (2022) examined cybersecurity implementation practices in organizations handling confidential digital information. The study found that organizations implementing layered security mechanisms such as authentication, access control, activity logging, and backup systems experience improved protection against unauthorized access and data breaches. The researcher emphasized that cybersecurity also involves monitoring, accountability, and risk management strategies.

Similarly, a related study conducted by William Stallings (2021) explored the implementation of security controls in web-based organizational systems. The researcher concluded that integrating authentication systems, access restrictions, activity monitoring, and backup procedures significantly improves information security and operational reliability.

Both studies are related to the proposed GreenLedger system because the proposed study also integrates multiple cybersecurity mechanisms intended to protect organizational records, payment transactions, and sensitive information within the organization.

References

Schneier, B. (2022). *Cybersecurity Practices in Organizational Information Systems*. Cybersecurity Review Journal.

Stallings, W. (2021). *Security Controls in Organizational Web Systems*. Information Security Journal.

2.3 Local Studies

2.3.1 Digital Records Management in the Philippines

Studies conducted in the Philippines emphasize the importance of digital records management systems in improving organizational efficiency, accessibility of information, and operational reliability. Local organizations transitioning from manual processes to computerized systems experience improvements in document management and transaction monitoring.

A study conducted by Maria Gamba, Guevara, and Naval (2019) examined the implementation of electronic records management systems in Philippine institutions. The researchers found that computerized systems improve document organization, retrieval efficiency, and operational reliability compared to traditional paper-based processes. The study also emphasized that digital systems reduce risks associated with misplaced records and improve accessibility of organizational information.

The study is related to the proposed GreenLedger system because both studies focus on improving organizational records management through computerized systems and centralized information storage.

Reference

Gamba, M., Guevara, R., & Naval, V. (2019). *Electronic Records Management in Philippine Institutions*. Philippine Journal of Librarianship.

2.3.2 Web-Based Management Systems in Philippine Organizations

A study conducted by John Paul Reyes, Santos, and Cruz (2021) evaluated the implementation of web-based management systems in small and medium-sized organizations in the Philippines. The researchers found that web-based systems improved operational efficiency, simplified transaction monitoring, and enhanced accessibility of organizational records.

The study emphasized that centralized web systems reduce dependency on manual processes and improve communication among organizational personnel. The researchers also noted that organizations using web-based systems experienced faster retrieval of information and improved monitoring of organizational activities.

This study is related to the proposed GreenLedger system because both studies focus on implementing web-based organizational systems intended to improve records management and operational processes.

Reference

Reyes, J. P., Santos, M., & Cruz, R. (2021). *Implementation of Web-Based Management Systems in Philippine Organizations*. Asia Pacific Journal of Information Technology.

2.3.3 Payment Monitoring Systems in the Philippines

A study conducted by Angela Dela Cruz (2022) explored the use of computerized payment monitoring systems in local businesses and service organizations in the Philippines. The study found that digital payment monitoring systems improved the

accuracy of financial records, reduced transaction delays, and simplified balance monitoring processes.

The researcher explained that organizations using automated payment systems experienced fewer computation errors and improved monitoring of installment-based transactions. The study also highlighted the importance of maintaining secure financial records through authentication and access control mechanisms.

The findings are relevant to the proposed GreenLedger system because the present study also focuses on secure and accurate payment tracking within organizational operations.

Reference

Dela Cruz, A. (2022). *Computerized Payment Monitoring Systems in Philippine Service Organizations*. Philippine Journal of Information Systems.

2.3.4 Information System Implementation in Philippine Organizations

A study conducted by Michael Anyayahan (2020) examined the implementation of organizational information systems in Philippine institutions. The researcher found that computerized systems improve operational productivity, reduce manual workloads, and support more efficient management of organizational records and transactions.

The study emphasized that digital systems improve coordination among departments and provide centralized access to organizational information. The researcher also highlighted that organizations adopting computerized systems become more capable of managing increasing volumes of records and operational activities.

This study is related to the proposed GreenLedger system because the proposed research also focuses on implementing an organizational information system intended to improve operational efficiency and secure management of records.

Reference

Anyayahan, M. (2020). *Information System Implementation in Philippine Organizations*. Philippine Information Technology Journal.

2.3.5 Security-Focused Systems in Philippine Schools and Businesses

A study conducted by Jerome Bautista, Ramos, and Villanueva (2023) evaluated the implementation of security-focused information systems in educational institutions and businesses in the Philippines. The researchers found that systems implementing authentication, access control, and activity logging mechanisms provided improved protection against unauthorized access and data misuse.

The study emphasized that organizations handling confidential information should prioritize cybersecurity measures to protect records and maintain operational reliability. The researchers also concluded that activity monitoring and user accountability significantly improve organizational security management.

This study supports the proposed GreenLedger system because the proposed study also incorporates cybersecurity mechanisms such as authentication, role-based access control, and audit logging to protect sensitive organizational information.

Reference

Bautista, J., Ramos, P., & Villanueva, K. (2023). *Security-Focused Information Systems in Philippine Organizations*. Journal of Philippine Cybersecurity Research.

2.3.6 Local Electronic Records Management Studies

A study conducted by Christine Villamor (2021) explored electronic records management practices in local Philippine organizations transitioning from manual systems to digital platforms. The study found that electronic records systems improve document accessibility, simplify information retrieval, and reduce operational inefficiencies associated with paper-based record management.

The researcher emphasized that digital systems improve long-term preservation of records and strengthen organizational accountability through centralized monitoring and documentation. The study also highlighted the importance of implementing backup systems and security mechanisms to protect digital records.

This study is related to the proposed GreenLedger system because the proposed study also focuses on centralized digital records management and organizational information security.

Reference

Villamor, C. (2021). *Electronic Records Management Practices in Philippine Organizations*. Philippine Journal of Administrative Systems.

2.4 Synthesis of Reviewed Literature

The reviewed literature and studies emphasize the importance of information systems, cybersecurity mechanisms, and digital records management in improving organizational operations and protecting sensitive information. The reviewed literature discussed how computerized systems contribute to operational efficiency, centralized data management, secure transaction monitoring, and organizational accountability. Concepts such as web-based systems, database management systems, authentication mechanisms, password hashing, audit logging, backup systems, and role-based access control provide important foundations for developing secure organizational platforms.

The reviewed foreign studies highlighted the effectiveness of secure information systems in improving records management, payment monitoring, and organizational security. Several studies emphasized the importance of implementing role-based access control, activity monitoring, authentication systems, and secure databases to protect organizational information against unauthorized access and operational risks. These studies demonstrated how computerized systems improve accessibility, efficiency, and accountability within organizational environments.

Similarly, the reviewed local studies showed that Philippine organizations increasingly adopt computerized systems to improve operational processes, records management, and transaction monitoring. Local studies emphasized the importance of centralized information systems, web-based platforms, and cybersecurity implementation in improving organizational productivity and protecting sensitive records. The studies also revealed that organizations transitioning from manual processes to digital systems experience improvements in efficiency, accessibility, and data management.

Despite the existence of various studies related to records management systems, payment monitoring systems, and cybersecurity implementation, limited studies focus

specifically on integrated systems designed for memorial park service organizations. Most reviewed studies focus on general organizational environments, educational institutions, or business operations rather than memorial park services involving long-term client records, burial lot management, and installment-based payment monitoring.

Therefore, the present study differs from previous studies by developing GreenLedger: A Role-Based Secure Records and Payment Management System for Memorial Park Services. The proposed system integrates records management, payment tracking, role-based access control, authentication mechanisms, activity logging, and backup systems within a centralized web-based platform intended specifically for memorial park service operations. Through these features, the study aims to improve organizational efficiency, strengthen data security, and support reliable management of sensitive client and financial records.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study utilizes a **Developmental research design** in the development of **GreenLedger: A Role-Based Secure Records and Payment Management System for Memorial Park Services**. Developmental research is commonly used in information technology and system development studies because it focuses on the design, development, implementation, and evaluation of computerized systems intended to address existing organizational problems and operational limitations.

The primary purpose of this research design is to develop a secure and efficient web-based system capable of improving the records and payment management processes of Greengarden Inc. The study specifically addresses the challenges encountered in the organization's current manual and spreadsheet-based processes, including difficulties in data organization, delayed retrieval of records, inconsistent payment monitoring, limited access control, and insufficient security mechanisms for handling sensitive client and financial information.

The developmental research design is appropriate for this study because the research does not merely aim to observe or describe organizational problems. Instead, it seeks to create an actual technological solution that can improve operational efficiency, strengthen cybersecurity practices, and support secure management of organizational records and payment transactions.

The study involves several interconnected processes including requirements gathering, system analysis, system design, software development, testing, deployment, and evaluation. Throughout these processes, the researchers will analyze the organization's operational workflow and develop a system that aligns with the identified functional and security requirements of Greengarden Inc.

Furthermore, the study incorporates cybersecurity principles and secure software development practices as part of the research design. Security mechanisms such as Role-Based Access Control (RBAC), authentication systems, password hashing, input validation, activity logging, and backup systems are integrated into the system to ensure confidentiality, integrity, and controlled accessibility of records and financial information.

The developed system will undergo testing and evaluation to determine whether it satisfies the required functionality, usability, reliability, efficiency, and security standards expected from a secure organizational management system. Feedback gathered from authorized users during system evaluation will also be used to identify necessary improvements and refinements prior to the final implementation of the proposed system.

Through the use of developmental research design, the researchers aim to provide Greengarden Inc. with a secure, organized, and efficient records and payment management platform capable of improving operational processes and supporting reliable management of sensitive organizational information.

3.2 System Development Methodology

This study utilizes the **Agile software development methodology** in the development of **GreenLedger: A Role-Based Secure Records and Payment Management System for Memorial Park Services**. Agile methodology is a modern software development approach focused on iterative development, continuous improvement, adaptability, and collaboration between developers and stakeholders throughout the development process.

Unlike traditional software development methodologies that follow rigid sequential phases, Agile methodology allows continuous revisions and improvements during development. This approach enables developers to identify system issues earlier, incorporate feedback continuously, and adapt to changing organizational requirements more effectively.

The Agile methodology is appropriate for this study because the proposed system involves multiple modules and security mechanisms that require continuous testing, modification, and enhancement. Since the system handles sensitive organizational and financial information, continuous evaluation and security improvement are essential throughout the development process.

The researchers selected Agile methodology because it supports flexibility, iterative testing, and progressive enhancement of system features. Through Agile development, the researchers can gradually improve the functionality, usability, and cybersecurity implementation of the proposed system based on the operational requirements of Greengarden Inc.

The Agile development process for GreenLedger consists of the following phases:

3.2.1 Requirements Analysis

The first phase of the Agile methodology involves gathering and analyzing information regarding the current operational procedures of Greengarden Inc. This phase focuses on identifying the organization's existing workflow, operational challenges, and system requirements.

The researchers will conduct interviews, observation, and document analysis to understand how client records, memorial lot information, and payment transactions are currently managed within the organization. Existing ledgers, spreadsheets, payment records, and operational documents will be reviewed to determine the required system features and database structure.

During this phase, the researchers will also identify cybersecurity concerns associated with the current manual and spreadsheet-based processes. Security-related requirements such as authentication mechanisms, role-based access control, activity monitoring, and backup systems will be analyzed to ensure that the proposed system addresses organizational security needs.

The requirements analysis phase serves as the foundation for system planning and development because it determines the functionalities, security controls, and operational features necessary for the proposed system.

3.2.2 System Design

After gathering the necessary requirements, the researchers will proceed to the system design phase. This phase focuses on planning the structure, workflow, database architecture, and security mechanisms of the proposed system.

The researchers will create several diagrams and models to visualize how the system will operate. These include:

- Data Flow Diagrams (DFD)
- Entity Relationship Diagram (ERD)
- Use Case Diagram
- System Architecture Diagram
- Flowcharts
- IPO Diagram

These diagrams help illustrate how data moves within the system, how users interact with system modules, and how different system components communicate with one another.

The system design phase also includes designing the user interface, database structure, user roles, access permissions, and security mechanisms. The researchers will ensure that the system design aligns with the operational requirements and cybersecurity objectives identified during the requirements analysis phase.

Special attention will be given to implementing secure design principles such as controlled access management, secure authentication, activity monitoring, and data protection mechanisms.

3.2.3 Development

The development phase involves the actual coding and implementation of the proposed system based on the approved system design.

The backend of the system will be developed using Python with the Django framework. Django is selected because it provides secure development features such as built-in authentication, password hashing, session management, and input validation mechanisms.

The frontend interface will be developed using:

- HTML
- CSS
- JavaScript

These technologies will be used to create responsive and user-friendly interfaces for managing client records, payment transactions, and administrative functions.

The system database will be implemented using SQLite for storing:

- Client records
- Memorial lot information
- Payment transactions
- User accounts
- Activity logs

During development, the researchers will continuously integrate cybersecurity mechanisms such as:

- Role-Based Access Control (RBAC)
- Authentication systems
- Password hashing

- Input validation
- Activity logging
- Backup systems
- Office-restricted access

Each module will be developed incrementally and tested regularly to ensure that system functions operate correctly and securely.

3.2.4 Testing

After development, the system will undergo several testing procedures to ensure functionality, reliability, usability, and security.

Testing is important to identify errors, vulnerabilities, and operational issues before deployment. The researchers will conduct various testing methods including:

- Unit Testing
- Integration Testing
- System Testing
- Security Testing
- User Acceptance Testing (UAT)

Each module will be evaluated to determine whether it performs according to the specified system requirements.

Security testing will also be conducted to verify the effectiveness of authentication mechanisms, access restrictions, password protection, and activity monitoring features.

Issues identified during testing will be corrected and improved before system deployment.

3.2.5 Deployment (Prototype)

Once the system passes testing and evaluation, the prototype will be deployed within the office environment of Greengarden Inc.

The deployment phase involves installing and configuring the system for organizational use. Authorized personnel including administrators, accounting staff, and general staff will be given access to the system according to their assigned roles and permissions.

The deployment phase allows users to evaluate the functionality, usability, and operational effectiveness of the proposed system within an actual organizational setting.

3.2.6 Continuous Improvement

Agile methodology emphasizes continuous improvement throughout the development process.

Feedback gathered from users during system testing and deployment will be analyzed to identify additional improvements, security enhancements, and usability modifications needed by the organization.

The researchers will continuously improve the system based on:

- User feedback
- Operational observations
- Security assessments
- System evaluation results

This phase ensures that the proposed system remains aligned with the operational and cybersecurity requirements of Greengarden Inc.

3.3 System Architecture

The GreenLedger Records and Payment Management System follows a client-server architecture utilizing a three-tier structure composed of the presentation layer, application layer, and data layer. This architectural design separates user interaction, system processing, and data storage to improve system organization, maintainability, scalability, and security.

The client-server architecture allows authorized users to access the system through a centralized web-based platform connected within the organization's designated office environment. This setup improves communication between users and the system while maintaining centralized control over organizational records and financial transactions.

The three-tier architecture is appropriate for the proposed system because it separates responsibilities among system components, making the application easier to manage, secure, and maintain.

3.3.1 Presentation Layer

The presentation layer serves as the user interface of the system. This layer allows users to interact with the application through web browsers using computers connected within the office environment of Greengarden Inc.

The frontend interface is developed using:

- HTML
- CSS
- JavaScript

The presentation layer provides users with interfaces for:

- User login and authentication
- Client information management
- Memorial lot management
- Payment recording and monitoring
- Activity log viewing
- Report generation
- Administrative functions

The interface is designed to be user-friendly, organized, and accessible for authorized organizational personnel. Different interfaces and functionalities are displayed depending on the assigned user role within the system.

For example:

- Administrators can manage users and activity logs.
- Accounting staff can manage payment transactions.
- General staff can manage operational records.

The presentation layer communicates user requests to the application layer for processing and retrieves processed information from the system database.

3.3.2 Application Layer

The application layer serves as the core processing component of the proposed system. This layer handles business logic, data processing, system validation, authentication procedures, and security enforcement.

The application layer is developed using Python with the Django framework.

Django is selected because it provides several advantages including:

- Secure authentication mechanisms
- Built-in password hashing
- Input validation support
- Session management
- Role-based access implementation
- Database integration
- Security protection against common web vulnerabilities

The application layer processes all requests submitted by users through the presentation layer. This includes:

- Validating login credentials

- Verifying user permissions
- Processing payment transactions
- Updating records
- Generating activity logs
- Managing backup procedures

The application layer also implements the cybersecurity mechanisms of the proposed system including:

Authentication System - Verifies user identity before allowing system access.

Role-Based Access Control (RBAC) - Restricts user access according to assigned organizational roles.

Password Hashing - Protects user passwords before storing them in the database.

Input Validation - Prevents invalid or malicious data entries.

Activity Logging - Records user actions and system activities.

Backup and Recovery - Supports protection against data loss incidents.

Through these mechanisms, the application layer ensures that the proposed system maintains security, accountability, and operational reliability.

3.3.3 Data Layer

The data layer is responsible for storing, retrieving, updating, and managing organizational information within the system.

The proposed system uses SQLite as its database management system.

The database stores important organizational data including:

- Client personal information
- Memorial lot records
- Payment transactions
- User accounts
- Activity logs
- System records

The database structure is designed to maintain data consistency, accessibility, and integrity.

The data layer supports secure communication with the application layer to ensure that only authorized and validated requests can access or modify stored information.

Security measures implemented within the data layer include:

- Controlled database access
- Data validation
- Password hashing storage
- Backup file generation
- Transaction consistency management

The centralized database structure improves organizational efficiency because records can be retrieved, updated, and monitored more efficiently compared to manual and spreadsheet-based processes.

3.4 Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Django Framework)
- Database: SQLite
- Development tools: Visual Studio Code

3.5 Data Gathering Techniques

- **Interviews** – Conducted with the manager of GreenGarden Inc. to understand the current workflow, challenges, and requirements for client records and financial accounting. This provided insights into what features are needed for the proposed system.
- **Document Analysis** – Reviewed existing ledgers, client records, and memorial lot plans provided by the manager. This helped determine the necessary data fields and informed the design of a secure and

3.6 Testing Procedures

- Unit Testing
- System Testing
- User Acceptance Testing

3.7 Evaluation Criteria

The system will be evaluated based on:

- Functionality
- Usability
- Security
- Efficiency
- Reliability

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

User → GreenLedger System → Database

Process Explanation:

- User logs in and submits data (records or payments)
- System processes and validates the input
- Data is stored in the database
- System retrieves and displays requested information

Entity Relationship Diagram (ERD)

Entities:

- **ClientPersonalInfo** (id, client_first_name, client_middle_name, client_last_name, client_address, client_contact_number, client_civil_status, client_date_birth, client_religion, client_occupation, client_employer_name, client_employer_address, client_spouse_name, client_spouse_date_birth, client_spouse_occupation, client_spouse_employer, client_id_type, client_id_number, client_date_issued, client_place_issued)
- **Beneficiary** (client, name, relationship)
- **ClientStatus** (id, client, plan, monthly_payment, duration, months_remaining, start_date, balance, paid_balance, date_paid, status)
- **Payment** (client_status, month, amount, is_paid, date_paid, ordering, status)
- **UserLog** (id, role, first_name, middle_name, last_name, date_of_birth, government_id, phone_number, address, email, emergency_contact_name, emergency_contact_number, time_in, time_out, activities, pin)

Relationships:

- User manages Client records
- Client makes Payment
- User generates Activity Logs

Use Case Diagram

Actors:

- Administrator
- Accounting Staff
- General Staff

Use Cases:

- Login / Logout
- Manage Users (Admin)

- Manage Client Records
- Record Payments (Accounting)
- View Reports
- View Activity Logs (Admin)